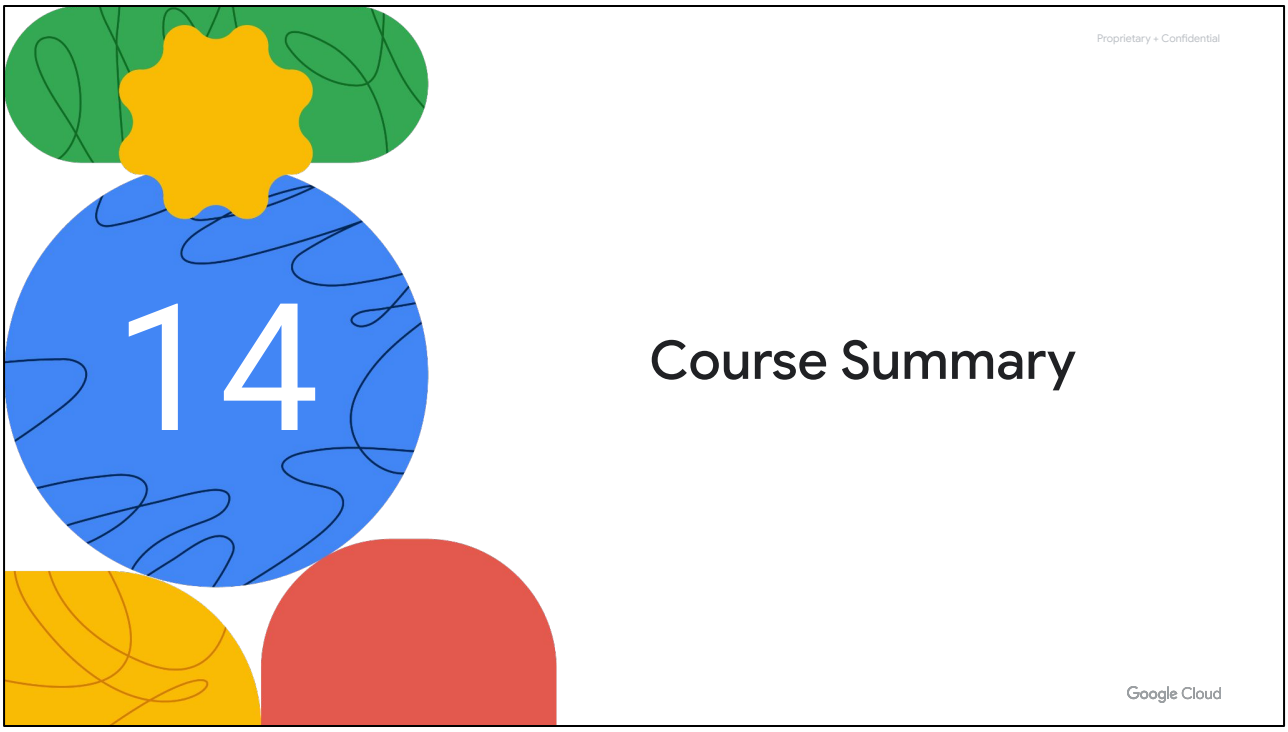

This brings us to the end of the “Manage Scalable Workloads in Google Kubernetes Engine ” course! We covered many topics, so let’s take a moment to review what you learned.

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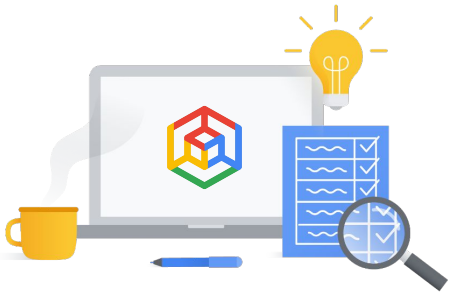
Module 1
Introduction to GKE at Scale



01 Multi-cloud and multi-cluster overview

02 Introduction to GKE Fleets

03 Sameness and trust

04 GKE fleet management features

Google Cloud

The first section of the was all about scaling you infrastructure in GKE

You learned about:

- Multi-cloud and multi-cluster requirements.
- GKE Fleets at a high-level.
- Sameness and trust.
- And GKE fleet management features.

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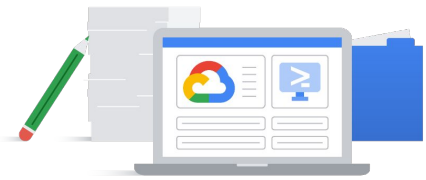
Module 2
Multi-cluster Architecture



01 Centralized cluster management

02 Multi-cluster GKE Architecture

03 Create, connect, and manage GKE fleet clusters

04 Access GKE fleet clusters

Google Cloud

Next, you learned how GKE can be used to centralize cluster management for multi-cluster environments, including:

- Centralized cluster management.
- Examining the architecture of Multi-cluster GKE
- Creating connecting, and managing GKE fleet clusters.
- And securely accessing GKE fleet clusters.

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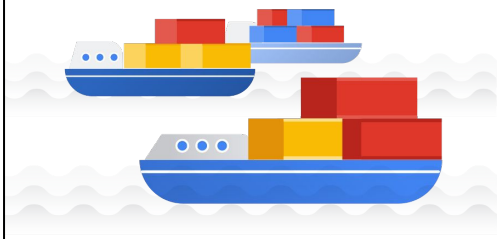
Module 3
Fleets and Teams



01 GKE fleets

02 Example fleet solutions

03 Fleet team management

04 Fleet management

Google Cloud

The third section of the course covered fleets and teams.

You explored:

- GKE fleets.
- Example fleet solutions.
- Fleet team management.
- And fleet management.

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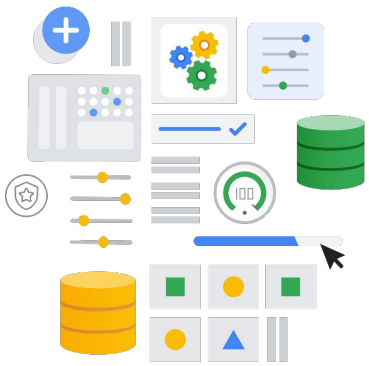
Module 4
Managing GKE Configuration at Scale



01 Configuration management challenges

02 Centralized configuration management at scale

03 Config Sync

04 Policy Controller

05 Config Connector

06 Blueprints

Google Cloud

From there, you moved on to managing configuration at scale, where you learned about:

- Configuration management challenges.
- Centralized configuration management at scale.
- Config Sync.
- Policy Controller.
- Config Connector.
- And Blueprints.

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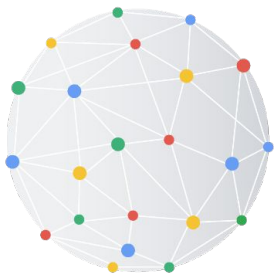
Module 5
Fleet Networking



01 Fleet networking communications

02 Pod discovery in GKE

03 Multi-cluster services

04 Configuring Multi-cluster services

05 Multi-cluster Gateway

06 Configuring Multi-cluster Gateways

Google Cloud

Then we journeyed further into the world of fleet networking in the fifth section, where you explored:

- Fleet networking communications.
- Pod discovery in GKE.
- Multi-cluster Services.
- Configuring multi-cluster Services.
- Multi-cluster gateway.
- And Configuring multi-cluster gateways.

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Module 6
Cloud Service Mesh

01 Introduction to Cloud Service Mesh

02 Provisioning Cloud Service Mesh

03 Handling requests

04 Cloud Service Mesh dashboards and support

Google Cloud

In the sixth section of the course, you explored Cloud Service Mesh.

The section covered:

- An introduction to Cloud Service Mesh.
- Provisioning Cloud Service Mesh.
- Handling requests.
- And Cloud Service Mesh dashboards and support.

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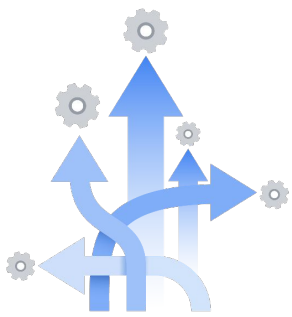
Module 7
Cloud Service Mesh Routing



01 Configuring Cloud Service Mesh with Istio API resources

02 Configuring VirtualService and DestinationRule

03 Configuring ServiceEntry

04 Configuring a Gateway

05 Configuring a WorkloadEntry and WorkloadGroup

06 Network resilience and testing

Google Cloud

In the seventh section of the course, you learned about Cloud Service Mesh routing. This section covered:

-
- Configuring Cloud Service Mesh with Istio API resources.
- Configuring VirtualService and DestinationRule.
- Configuring ServiceEntry.
- Configuring a Gateway.
- Configuring a WorkloadEntry and WorkloadGroup.
- And network resilience and testing.

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Module 8
Service Mesh Security

01 Authentication and encryption

02 Service authentication in the mesh

03 End-user authentication in Cloud Service Mesh

04 Authorization in Cloud Service Mesh

Google Cloud

From there, you learned all about service mesh security, including:

- Authentication and encryption.
- Service authentication in the mesh.
- End-user authentication in Cloud Service Mesh.
- Authorization in Cloud Service Mesh.

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Module 9
Multi-cluster Networking with Cloud Service Mesh

01 Single network east-west routing

02 Multiple network east-west routing

Google Cloud

The ninth section of the course covered multi-cluster networking with Cloud Service Mesh.

You explored single network east-west routing and multiple network east-west routing.

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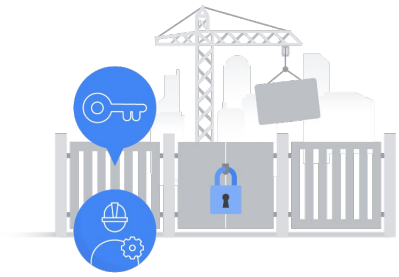
Module 10
Manage Identity in GKE with Authentication



01 Introduction to GKE Identity Service

02 Connect gateway overview

03 Configuring connect gateway for authentication and authorization

04 Accessing clusters with GKE Identity Service

05 Authenticating third-party identities with GKE Identity Service

06 Fleet Workload Identity

Google Cloud

From there, you moved on to managing identity in GKE with authentication, where you learned about:

- GKE Identity Service.
- Connect gateway.
- Configuring connect gateway for authentication and authorization.
- Accessing clusters with GKE Identity Service.
- Authenticating third-party identities with GKE Identity Service.
- And fleet Workload Identity.

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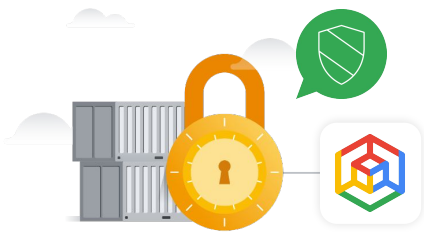
Module 11
Security Posture, Compliance, and Preventative Controls



01 GKE security posture overview

02 Security posture dashboard

03 Implementing node security

04 Vulnerability scanning

05 Additional security services

Google Cloud

Next, you learned about security posture, compliance, and preventative controls, including:

- An overview of GKE security posture.
- The security posture dashboard.
- Implementing node security.
- Vulnerability scanning.
- And additional security services.

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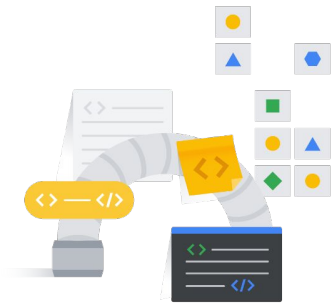
Module 12
CI/CD at Scale



01 CI/CD in Google Cloud

02 Cloud Build and GKE

03 Cloud Deploy and GKE

04 Cloud Deploy: Policies,Deployments and Security

05 Cloud Run and Knative Serving

06 Cloud Deploy and Knative Serving

07 CI/CD in a private network

08 Securing the software supply chain

Google Cloud

In section twelve, you learned about CI/CD at Scale. This section covered:

- CI/CD in Google Cloud.
- Cloud Deploy and GKE.
- Cloud Deploy and Cloud Run.
- CI/CD in a private network.
- And securing the software supply chain.

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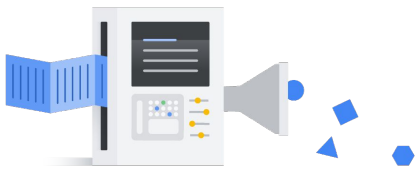
Module 13
GKE and AI



01 AI and GKE overview

02 AI model training on GKE

03 AI model serving on GKE

04 AI cost management on GKE

Google Cloud

Finally, in the final section of the course you learned about the relationship between GKE and AI. This section covered:

- An overview of AI and GKE.
- AI model training on GKE.
- AI model serving on GKE.
- AI cost management on GKE.

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We'd love your feedback!

Please take a few minutes to complete the survey and help us improve this course for future learners.

Survey

Google Cloud

Your feedback matters and is greatly appreciated! Please share your thoughts on your experience by completing a short survey. It only takes a few minutes and will help us improve future sessions.

[If virtual] I've also added the link in the chat.

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Thank you

Google Cloud

This brings up to the end of this course! Thank you for your time and participation. See you next time!